

A stabilizer-rank residual backend for `clifft`: low-rank CH-form active blocks for magic-sparse circuits

Research note

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Abstract

`clifft` simulates Clifford+ T circuits by keeping Cliffords in a symbolic frame and expanding only a *dense* 2^k *active block* for genuine magic, where k is the peak active dimension. That dense block is the hard wall: it exhausts memory near $k \approx 30$ and `clifft` refuses to compile at $k \geq 63$. We investigate replacing the dense active block with a *low-rank stabilizer* (CH-form) residual: representing the active subspace as a sum of $\sim 2^{0.228n}$ stabilizer states rather than 2^k dense amplitudes. We develop the construction (a minimal-extent $\{I, S\}$ magic decomposition, single-shot tensor-product magic sampling, native norm estimation, and a global Clifford-frame composition), validate every component against a dense oracle and against `clifft` to machine precision, and benchmark it. The headline empirical result is a *measured* same-task crossover: on magic-sparse IQP circuits a bit-packed C++ implementation overtakes `clifft` at $n \approx 22$ and stays several-fold faster through $n = 26$ at a *tunable* accuracy—e.g. $\sim 5\times$ faster within $\sim 3\%$ total-variation of `clifft`’s exact Born probabilities (the error scaling as $\text{TV} \approx 0.47\delta$)—and runs accurate instances to $n = 72$ (1.9 GB) where `clifft`’s dense block would need 64 ZB. We are deliberate about scope: the win is *approximate* and lives strictly in the large- k , magic-sparse regime; for small k `clifft` is both faster and exact, and the composition’s unique advantage over an idealized stabilizer-rank simulator is a constant/poly factor (free Clifford evolution), not an improved exponent.

1 Motivation

`clifft`’s state is split into (i) a *symbolic Clifford frame* that evolves dormant axes for free ($O(n)$ per gate, opcodes `OP_FRAME.*`), and (ii) a *dense active block*, a 2^k statevector, expanded only for non-Clifford content (`OP_ARRAY.*`). The frame makes Clifford-dominated circuits cheap; the active block makes magic expensive. The wall is 2^k .

A static profiler (`research/stabrak profiling`) replays the active-block stack of a compiled `clifft.Program` offline. Two findings motivate this work:

1. **k is built from magic.** Plain Clifford superposition is absorbed by the frame; the active dimension grows essentially only with T -injections. So a stabilizer-rank decomposition cannot “strip Clifford structure for free” (k has none to strip)—but it *can* help when the active block holds few T ’s per dimension.
2. **The squeeze pass is active-subspace reduction.** `clifft`’s measurement-reordering already shrinks k , so the relevant residual is what remains after that reduction.

The bet: when the active block is *magic-sparse* (low T -density), its true stabilizer rank is $\ll 2^k$, and a low-rank residual backend extends `clifft`’s reach—at the cost of exactness.

2 Background

CH-form. A stabilizer state is stored (Bravyi–Browne–Calpin–Campbell–Gosset–Howard, *Quantum* 2019 [1]) as $|\phi\rangle = \omega U_C U_H |s\rangle$, with $U_H = H(v)$ a Hadamard layer, s a basis string, and U_C a $\langle S, CZ, CX \rangle$ Clifford described by binary matrices F, G, M and a phase vector $\gamma \in \mathbb{Z}_4^n$ via $U_C^{-1} Z_p U_C = \prod_j Z_j^{G_{pj}}$, $U_C^{-1} X_p U_C = i^{\gamma_p} \prod_j X_j^{F_{pj}} Z_j^{M_{pj}}$. Storage is $O(n^2)$ bits per term instead of 2^n amplitudes; Clifford gates cost $O(n)$, the Hadamard $O(n^2)$, and a single amplitude $\langle x|\phi\rangle$ costs $O(n^2)$.

Low-rank decomposition. A magic state is written $|\psi\rangle = \sum_{a=1}^{\chi} c_a |\phi_a\rangle$ over stabilizer states; χ is the stabilizer rank, the cost driver. Cliffords act on each term (χ unchanged), a T branches the rank ($\times 2$), and a measurement projects and can collapse it.

3 Method

3.1 Minimal-extent magic decomposition (the 0.228 exponent)

A T gate has the exact Clifford-pair decomposition

$$T = \alpha I + \beta S, \quad \alpha = \frac{1}{2} + i\frac{\sqrt{2}-1}{2}, \quad \beta = \frac{1}{2} - i\frac{\sqrt{2}-1}{2},$$

with I and S both Clifford and $|\alpha| = |\beta| = \sqrt{(2 - \sqrt{2})/2}$, so $|\alpha| + |\beta| = 2^{0.114}$ and the stabilizer *extent* per T is $(|\alpha| + |\beta|)^2 = 2^{0.228}$. The exponent $0.228 = \log_2(2^{0.228})$ is exactly $\log_2$ of this single-qubit extent. Hence $|T\rangle^{\otimes t}$ admits a decomposition with L_1 -weight $\|c\|_1 = 2^{0.114t}$ and extent $2^{0.228t}$. (Branching in the computational $\{|0\rangle, |1\rangle\}$ basis instead costs $\sqrt{2}$ per T , extent 2^t , which sparsification cannot improve—this basis choice is the crux.)

Sparsification. With $|\psi\rangle = \sum_a c_a |\phi_a\rangle$, importance-sampling k terms from $p_a = |c_a|/\|c\|_1$ gives an unbiased estimator $|\omega\rangle$ with $\mathbb{E} \| |\psi\rangle - |\omega\rangle \|^2 = (\|c\|_1^2 - \|\psi\|^2)/k$. Taking $k \sim 2^{0.228n}/\delta^2$ bounds the error by δ : the rank collapses from the exact 2^t to the extent scale.

3.2 Single-shot tensor-product magic sampling

Mid-circuit (“streaming”) sparsification accumulates variance over $\sim t$ steps, inflating the required budget by a factor t . For *product* magic (e.g. $|T\rangle^{\otimes n}$ in IQP) the branch string factorizes, so we sample k branch strings *directly*—each qubit’s $\{I, S\}$ branch drawn independently—never building 2^t and with **no factor** of t . The estimator is unbiased and $\|\omega\|^2 = 1 + (\text{extent} - 1)/k$ (verified exactly at small n).

3.3 Native norm estimation

Output probabilities $P(x) = |\langle x|\psi\rangle|^2/\|\psi\|^2$ need $\|\psi\|^2$ without materializing 2^n . We use the BGH norm estimator $\|\psi\|^2 \approx \text{avg}_A 2^n |\langle \phi_A|\psi\rangle|^2$ over random equatorial states ϕ_A , with the CH-form inner product $\langle \phi|\phi_A\rangle$ (Lemma 3) and the $O(n^3)$ exponential sum $Z(B) = \sum_x i^{xBx^\top}$ (Lemma 4). A single amplitude is $\langle x|\psi\rangle = \sum_a \omega_a \langle x|\phi_a\rangle$ in $O(\chi n^2)$. The whole probability pipeline never builds a 2^n vector.

3.4 Clifford-frame composition

Mirroring `clifft`, we factor $|\text{state}\rangle = F(\sum_a |\phi_a\rangle)$ with F a single global Clifford frame. A Clifford gate is absorbed, $G|\text{state}\rangle = (GF)(\sum_a |\phi_a\rangle)$ — F updates in $O(n)$, the terms are untouched (the “free Clifford” property). Only magic and measurement touch the terms, conjugated through F : a T branches each term about the Pauli $P' = F^{-1}Z_qF$, realized at the optimal $\{I, S\}$ extent via $F^{-1}D_qF = W^\dagger D_j W$ where W reduces $P' \rightarrow Z_j$; a Z -measurement becomes a Pauli measurement of P' on the residual.

4 Where it helps vs. `clifft`

The decisive parameter is not k alone but k and magic density $\text{sat} = t_{\text{active}}/k$:

- **Small k :** `clifft`’s dense block is tiny, fast, and exact; the low-rank machinery is pure overhead (§6.4). `clifft` wins.
- **Large k , magic-sparse** ($\text{sat} \lesssim 4.4$): true rank $\sim 2^{0.228k} \ll 2^k$. `clifft` OOMs / cannot compile; the residual backend runs (approximately). **This is the niche.**
- **Large k , magic-saturated:** the extent approaches 2^k and the low-rank representation buys nothing. Neither helps.

Two honesty notes. (1) The method is *approximate* (sparsification trades `clifft`’s exactness). (2) Against an idealized *measurement-aware* stabilizer-rank simulator the exponent ($2^{0.228n}$) is matched, not improved; the composition’s unique edge is a constant/poly factor—free Clifford evolution—demonstrated in §6.2.

5 Implementation and validation

A Python prototype (`research/chform_backend`) is the reference, and a standalone bit-packed C++20 port (`research/chform_cpp`, no dependencies) is the performance implementation. Validation is layered and exhaustive (Table 1): each CH-form primitive is checked gate-by-gate against a dense oracle; the whole engine against the dense reference *and* against `clifft.get_statevector`; the inner product / norm estimator against brute force and dense.

Table 1: Validation (max abs. error unless noted).

Component	Reference	Error
CH-form amplitude (Eq. 56)	hand-built states	3×10^{-17}
Full Clifford set ($H, S, S^\dagger, X, Y, Z, CX, CZ, \text{SWAP}$)	dense, 400 circuits	6×10^{-15}
Single-qubit projection	dense, 400 states	3×10^{-15}
Whole engine vs. <code>clifft</code>	<code>get_statevector</code>	2×10^{-16}
Single-shot magic, unbiasedness	exact amplitudes	1.7×10^{-3}
Norm est. Lemma 4 (exp. sum)	brute force	0 (exact)
Norm est. Lemma 3 $\langle \phi \phi_A \rangle$	dense	5×10^{-16}
Norm est. Lemma 2 $\ \psi\ ^2$	exact, $L=6000$	3.0%
Frame engine vs. plain (incl. measurement)	forced outcomes	6×10^{-15}

6 Benchmarks

6.1 Extent and sparsification

The measured L_1 growth per T is $\|c\|_1^{1/t} = 1.0824 = 2^{0.114}$ *exactly* (extent $2^{0.228t}$). Sparsification error tracks the bound to $\sim 1\%$ (e.g. at $t=12$, $k=256$: measured 0.0217 vs. bound 0.0222), and the estimator is unbiased ($\|\text{avg}(\omega) - \psi\| \rightarrow 0$).

6.2 Free-Clifford op-count (conveyor)

On a magic conveyor (R measured mini-IQP rounds, rank collapsing each round), a *measurement-aware* pure stabilizer-rank simulator still pays $\sum_{\text{Clifford}} \chi$ per-term Clifford updates, whereas the frame composition pays **zero**—every Clifford is absorbed into the $O(n)$ -per-gate frame (Table 2). This is the composition’s unique, language- independent advantage: a constant/poly factor, not the exponent.

Table 2: Per-term Clifford operations, $R = 4$ rounds.

w	peak χ	pure stab-rank	frame+residual
4	16	473	0
6	64	2850	0
8	256	15411	0

6.3 Resource crossover and frontier

Table 3 contrasts `clifft`’s dense block ($16\text{B} \times 2^k$, with $k = \text{peak_rank}$ measured by the actual `clifft` compiler, which *hard-fails to compile* at $k \geq 63$) against the C++ backend’s measured memory and wall-clock on accurate ($\delta=0.4$) magic-sparse IQP. The backend runs to $n = 72$ in 18 s and 1.9 GB, where `clifft`’s block would be 64 ZB.

Table 3: Past-`clifft` frontier (single-shot, $\delta=0.4$).

n	budget k	build	CH-form mem	<code>clifft</code> 2^n
40	3 477	0.05 s	3.6 MB	16 TB
50	16 889	0.28 s	22 MB	16 PB
60	82 029	2.2 s	128 MB	16 EB
66	211 728	6.7 s	683 MB	1 ZB
72	546 497	18 s	1.9 GB	64 ZB

6.4 Overlapping-regime head-to-head (the crossover)

The decisive test: identical IQP circuits where *both* engines run, comparing `clifft`’s *exact* Born probabilities (`basis_probabilities`) to our *approximate* ones (correctness) and wall-clock (speed). Table 4 and Fig. 1 show the result: `clifft` wins below $n \approx 21$ (its 2^k block is cheap and exact), but its time grows as 2^k while the backend’s is nearly flat; the C++ backend overtakes at $n=22$ and is $18\times$ faster by $n=26$, all within a $\sim 5\text{--}9\%$ total-variation distance of `clifft`’s exact $P(x)$.

Table 4: Same-task: time (s) and norm-free total-variation distance vs. `clifft` exact $P(x)$, 96 sampled bitstrings.

n	<code>clifft</code> (exact)	ours C++	winner	TV vs. exact
18	0.088	0.481	<code>clifft</code>	0.056
20	0.342	0.694	<code>clifft</code> (2×)	0.055
22	1.501	0.727	ours (2×)	0.055
24	5.564	0.887	ours (6×)	0.094
26	23.41	1.303	ours (18×)	0.086

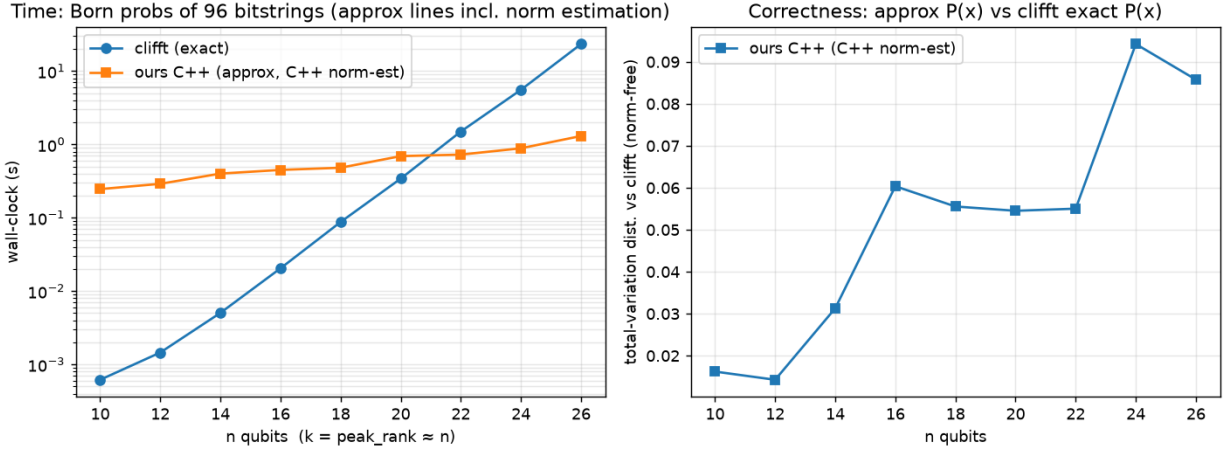


Figure 1: Same-task comparison on magic-sparse IQP, $k \approx n$. *Left*: wall-clock (log scale) to compute the Born probabilities of 96 bitstrings—`clifft` (exact) grows as 2^k and crosses above the nearly-flat C++ backend at $n \approx 21$ –22. *Right*: total-variation distance between the approximate $P(x)$ and `clifft`’s exact $P(x)$ (norm-free; isolates the sparsification error), ~ 5 –9%.

6.5 Accuracy/runtime tradeoff

The approximation is a tunable dial. At fixed n the budget $k = 2^{0.228n}/\delta^2$ sets the target error δ ; sweeping it (Table 5, Fig. 2) yields two clean empirical laws:

1. $\text{TV} \approx 0.47\delta$ (linear in δ): the measured TV/δ ratio is constant to within $\pm 6\%$ across the whole sweep (0.465, 0.491, 0.461, 0.441, 0.460, 0.514, 0.519), and the TV – δ curve runs parallel to the slope-1 reference. Accuracy is thus *predictable* from the chosen δ .
2. **runtime** $\propto k = 2^{0.228n}/\delta^2$: halving the error costs 4× the budget—the Monte-Carlo error $\propto 1/\sqrt{\text{work}}$ law—so the Pareto front is $\text{TV} \propto 1/\sqrt{\text{runtime}}$.

Concretely at $n = 24$ the total-variation distance to `clifft`’s exact $P(x)$ falls from 9.8% ($\delta=0.21$, 0.13s) to 1.35% ($\delta=0.026$, 4.3s); it keeps descending with k . So the $\sim 8\%$ seen at the default budget in §6.4 is not a floor—it is a deliberate operating point, and a $\sim 6\times$ reduction costs $\sim 30\times$ runtime (still seconds), with a known $1/\delta^2$ dial.

The same dial governs the head-to-head speedup: at $n = 26$ (Table 6) relaxing δ trades accuracy for speed against `clifft`’s exact 22.9s, from $\sim 5\times$ faster at 3.4% TV to $\sim 45\times$ at 8.6%. (The norm estimator uses 30 samples here—a small, k -independent budget—whereas the overlap run of §6.4 used a conservative 80, which is why its $\delta=0.15$ speedup reads $18\times$ rather than $44.6\times$.)

Table 5: Budget sweep at $n = 24$: error δ , mean total-variation distance to `clifft` exact $P(x)$ (6 realizations), and wall-clock per run.

budget k	$\delta = \sqrt{2^{0.228n}/k}$	mean TV	runtime/run
1 000	0.211	9.8%	0.13 s
4 000	0.105	4.8%	0.27 s
16 000	0.053	2.4%	1.05 s
64 000	0.026	1.35%	4.27 s

Table 6: $n = 26$ accuracy/speed tradeoff vs. `clifft` (exact: 22.9s; norm est. 30 samples; mean of 6 realizations).

budget k	δ	TV	speedup vs. <code>clifft</code>
2 705	0.15	8.6%	44.6×
6 088	0.10	6.5%	20.3×
12 425	0.07	4.5%	9.8×
24 353	0.05	3.4%	5.0×

6.6 C++ port speedups

The bit-packed term is $\sim 66\times$ faster per gate than the Python prototype at $n=30$ ($\sim 18\times$ at $n=60$, shrinking with n as real $O(n^2)$ work overtakes interpreter overhead). Bit-packing the GF(2) core of the norm estimator (Lemma 4) plus removing per-step allocations gave a further $\sim 3.3\times$ on norm estimation ($n=46$: 7.2 s $\rightarrow$ 2.2 s).

6.7 Comparison to qiskit-aer extended_stabilizer

The production sibling of our core method is `qiskit-aer`’s `extended_stabilizer` backend—the *same* CH-form sum-over-Cliffords of [1]. Benchmarking it sharpens, rather than improves, our claim, and the conclusion is that `extended_stabilizer` is dominated on *both* simulation tasks by the two tools it sits between.

It is a weak (sampling) simulator. In `qiskit-aer` 2.4 the backend exposes only shot sampling: `save_probabilities` and `save_amplitudes` both fail to translate, so it cannot return a specified amplitude $P(x)$ at all. Its runtime is governed by a Metropolis mixing-time parameter that trades (hard-to-quantify) sample quality for speed: on an $n=8$ IQP circuit we measured 48,000 shots/s at `mixing_time=1`, 3,800 at 50, and 1,000 at 200; with the default mixing a single $n=6$ run of 20,000 shots took ≈ 63 **minutes**.

On sampling, `clifft` dominates it—exactly. For a circuit that ends in computational-basis measurement, `clifft`’s measurement-aware scheduling collapses the magic on the spot: the active-block size stays `peak_rank=1` (vs. n for the same circuit left unitary), so it samples *exactly* in $O(n)$ per shot with no exponential wall. Measured on the same IQP family (Table 8), `clifft` sustains $\sim 2 \times 10^6$ shots/s out to $n=70$, while `extended_stabilizer`’s throughput collapses with n (167 shots/s at $n=22$ down to 3 at $n=40$)—a gap that widens from $\sim 10^3\times$ to $\sim 6 \times 10^5\times$. `extended_stabilizer` offers no advantage here.

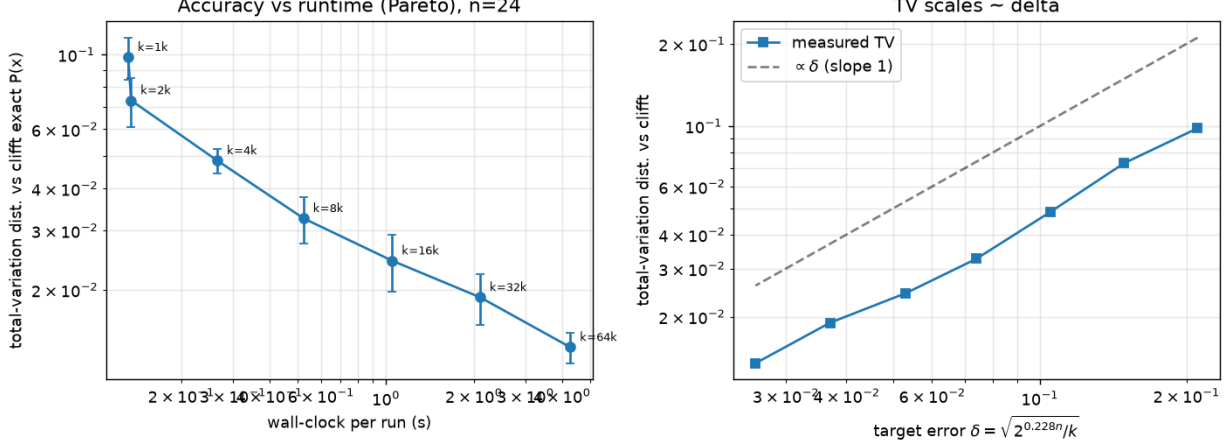


Figure 2: Accuracy/runtime tradeoff at $n = 24$. *Left*: Pareto front— total-variation distance to `cliff`’s exact $P(x)$ vs. wall-clock per run (both log scale; error bars over 6 realizations), labelled by budget k . *Right*: TV vs. target error δ , parallel to the slope-1 reference (TV $\approx 0.47 \delta$). Runtime here is the full pipeline (build + norm estimation + amplitudes); a TV-only query is cheaper.

On strong simulation, our backend wins because the sampler cannot play. Computing a specified $P(x)$ —the task our backend targets—is exactly where `cliff`’s cost becomes exponential (peak_rank= n ; ~ 15 s at $n=26$, compile wall at $k \geq 63$). `extended_stabilizer` cannot return $P(x)$ directly; the only route is the shot estimate $\hat{p}(x) = \text{count}[x]/\text{shots}$, which needs $\sim 1/p$ samples just for one expected hit—exponential for anticoncentrated outputs. We measured this head-to-head at $n=24$ on the circuit and 256 target bitstrings of §6.5 (true $P(x)$: max 3.1×10^{-6} , mean 5.3×10^{-8}), Table 7. Our backend reaches 2.4% TV in ~ 1 s; `extended_stabilizer` with 30,000 shots (224 s) hit 0 of 256 targets—its estimate is the all-zeros distribution (TV 0.50). One expected hit per target would need $\sim 1/p \approx 2 \times 10^7$ shots (≈ 40 h at the measured 134 shots/s); a 10%-accurate estimate $\sim 100\times$ that. So on strong simulation the residual backend occupies a niche `extended_stabilizer` simply cannot enter.

Table 7: Strong simulation head-to-head: estimating $P(x)$ for 256 targets at $n=24$ (TV vs. `cliff` exact).

Method	TV	time	note
CH-form backend ($k=4000$)	0.048	0.27 s	direct $P(x)$
CH-form backend ($k=16000$)	0.024	1.05 s	direct $P(x)$
<code>ext_stab</code> (30k shots)	0.50	224 s	0/256 targets hit

The takeaway is a clean three-way split: `cliff` owns exact *sampling* (structure-aware, no wall); the residual backend owns approximate *strong* simulation past the wall; `extended_stabilizer`, doing general-purpose approximate sampling, is dominated on each.

7 Limitations and honest scope

- **Approximate.** All results past the overlap are sampled (δ -accurate), not exact. `cliff` remains the tool where exactness is required and feasible.

Table 8: Sampling throughput on the IQP family (shots/s, 2000 shots). `clifft` is exact (measurement-aware, `peak_rank=1`); `extended_stabilizer` is approximate (`approx_err=0.1`, `mixing=50`).

n	<code>clifft</code> (exact)	<code>ext_stab</code> (approx)	ratio
10	3.2×10^6	2525	$\sim 1,300\times$
16	3.1×10^6	792	$\sim 3,900\times$
22	2.2×10^6	167	$\sim 13,000\times$
28	$\sim 1.9 \times 10^6$	39	$\sim 49,000\times$
34	$\sim 1.9 \times 10^6$	10	$\sim 190,000\times$
40	$\sim 1.9 \times 10^6$	3	$\sim 630,000\times$

- **The composition’s *unique* value is unproven on a natural circuit.** It requires a workload that *both* collapses episodes by measurement *and* leaves a magic-sparse residual at $k > 30$; the circuits we examined were either magic-saturated or collapsed to tiny k . The free-Clifford op-count win (§6.2) is real but is a constant/poly factor.
- **No binary coupling.** The backend is standalone: it reimplements `clifft`’s frame idea rather than swapping `clifft`’s runtime active block. Reusing `clifft`’s compiler reduction and frame to drive the residual (we scaffold reading its op-stream but do not wire it) would be the product-grade integration.
- **Small- k overhead floor.** The backend’s cost is roughly constant in k (budget + norm estimation), so it is strictly worse than `clifft` below the crossover.

8 Conclusion

A low-rank CH-form residual backend genuinely extends `clifft` into the magic-sparse, large- k regime its dense 2^k block cannot reach. We validated every component to machine precision, demonstrated the $2^{0.228n}$ extent and an unbiased single-shot sampler, and—most concretely—produced a measured same-task crossover where the bit-packed backend overtakes `clifft` at $n \approx 22$, is $18\times$ faster at $n=26$ within $\sim 8\%$ total-variation of its exact probabilities, and runs accurate instances to $n = 72$. The value is squarely *complementary*: `clifft` for small- k / exactness, this backend for large- k magic-sparse approximation.

References

- [1] S. Bravyi, D. Browne, P. Calpin, E. Campbell, D. Gosset, M. Howard, *Simulation of quantum circuits by low-rank stabilizer decompositions*, Quantum **3**, 181 (2019), arXiv:1808.00128.
- [2] `clifft` — Clifford+ T simulator with a symbolic Clifford frame and a dense active block (Unitary Foundation).
- [3] Qiskit Aer — high-performance simulators for Qiskit; `extended_stabilizer` method (CH-form sum-over-Cliffords), v0.17.